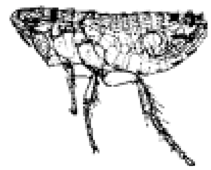
flea
NEWS 90 *November 2022*



http://esanetworks.org/

FLEA NEWS is a twice-yearly newsletter about fleas (Siphonaptera). Before including in publications any of the citations given here, please review for accuracy and correct formatting. Many of our sources are abstracting journals and current literature sources, and editors have not necessarily checked citations.

Recipients are urged to contribute items of interest to the profession for inclusion herein, including: New flea species descriptions, Flea research citations from journals or conference proceedings that are not available or indexed, Announcements and Requests for material, Contact information for a Directory of Siphonapterists (name, mailing address, email address, and areas of interest - Systematics, Ecology, Control, etc.), Abstracts of research planned or in progress, Book and Literature Reviews, Biography, Hypotheses, and Anecdotes. Send to:

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Contents

Editorial

"Inventorying parasites"

"I saw that for many years I had lost my way. I had been led astray by false trails and had been trying to do things contrary to my nature. I resolved to return to my childhood visions of studying nature and trying to protect wild things of the earth... We must follow our path with the courage of our convictions, refusing to be diverted by the clamour and confusion of less important demands."

- Ornithologist Margaret Nice, quoted in Simonds, pg. 194, 196.

Dear *Flea News* Reader,

We could do better biotic inventories simply by collecting and recording fleas and other parasites. We could acquire samples if ectoparasitologists collected endoparasites simultaneously, and vice versa. Carlson et al. (2020) advised that many field scientists:

> "lack the expertise or training to collect parasites without the support of a parasitologist. By working together, field parasitologists, zoologists, and curators could create and distribute protocols for collecting free-living species in ways that maximize collection and preservation of associated parasite data.... An exemplary document was recently produced for mammalogists, which describes the technical aspects of parasite collection in field studies for people with little to no parasite expertise (Galbreath et al., 2019) [see abstract below]. Similar documents are needed for all major vertebrate groups, especially amphibians and reptiles, which may have some of the most co-threatened parasites, but are severely under-represented in collections."
>
> "Another key priority will be to survey rare and threatened host species for parasites. These parasites might face the highest coextinction risk and might be the most poorly described and poorly studied symbiont fauna, given host rarity. Veterinarians often investigate host health, but rarely consider coextinction, so routine health examinations could be expanded to include parasite taxa not considered to threaten host health and to deposit samples in museum collections. Parasite data from these rare host species could be acquired when hosts are handled or permanently collected, or their blood, tissues, or feces are sampled. In many cases, this would not require additional permits beyond those already required for collecting free-living hosts. Furthermore, for animals that are being sacrificed and for rare animals (e.g., anesthetized wild cats that are being radio-collared), collecting parasite data would help to meet the ethical mandate to maximize scientific value stemming from each disturbance.

> Therefore, animal ethics and welfare committees and permitting agencies for wildlife trapping and collection can request that investigators working with rare and endangered species explain how their samples will be permanently preserved. For specimens of particular importance, permit-grantors could recommend that samples be archived in museum collections..., rather than laboratory freezers and cabinets, where they are inaccessible to the research community and may never be used again." (Carlson et al. 2020).

Most flea species are known from only a small number of collections, often only one collection! Additional collections and focused studies would deepen our knowledge on the faculties of fleas, and help conserve the wild hosts, communities, and ecosystems of fleas and other parasites beyond that of what we are capable of currently.

Yours in fleas,
R.L. Bossard
Editor, *Flea News*

Literature

Carlson, C.J., S. Hopkins, K.C. Bell, J. Doña, S.S. Godfrey, M.L. Kwak, K.D. Lafferty, M.L. Moir, K.A. Speer, G. Strona, M. Torchin, C.L. Wood. 2020. A global parasite conservation plan. Biol. Cons. 250:108596.

Simonds, M. 2022. Woman, Watching: Louise de Kiriline Lawrence and Songbirds of Pimisi Bay, ECW Press, Ontario, Canada.

Galbreath, K.E. et al. 2019. Building an integrated infrastructure for exploring biodiversity: field collections and archives of mammals and parasites. J. Mammalogy 100(2):382–393.

Abstract: Museum specimens play an increasingly important role in predicting the outcomes and revealing the consequences of anthropogenically driven disruption of the biosphere. As ecological communities respond to ongoing environmental change, host–parasite interactions are also altered. This shifting landscape of host–parasite associations creates opportunities for colonization of different hosts and emergence of new pathogens, with implications for wildlife conservation and management, public health, and other societal concerns. Integrated archives that document and preserve mammal specimens along with their communities of associated parasites and ancillary data provide a powerful resource for investigating, anticipating, and mitigating the epidemiological, ecological, and evolutionary impacts of environmental perturbation. Mammalogists who collect and archive mammal specimens have a unique opportunity to expand the scope and impact of their field work by collecting the parasites that are associated with their study organisms. We encourage mammalogists to embrace an integrated and holistic sampling paradigm and advocate for this to become standard practice for museum-based collecting. To this end, we provide a detailed, field-tested protocol to give mammalogists the tools to collect and preserve host and parasite materials that are of high quality and suitable for a range of potential downstream analyses (e.g., genetic, morphological). Finally, we also encourage increased global cooperation across taxonomic disciplines to build an integrated series of baselines and snapshots of the changing biosphere.

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Announcements

The journal *Diversity* preparing issue on "Epidemiology, Evolution, Phylogeny and Taxonomy of Siphonaptera"

"Dear Colleagues,

Fleas (Siphonaptera) constitute a highly distinct group of holometabolous bloodsucking insects which currently include about 2600 species-level taxa belonging to 16 families and 238 genera. Some authors have argued that Siphonaptera is the most completely studied order of insects, and although this is perhaps true from a morphological classification perspective, from a phylogenetic standpoint they have been sorely neglected as a group. Classically, the major obstacle in flea phylogeny has been their extreme morphological specialization associated with ectoparasitism, and the inability of systematics to homogenize characters adequately across flea and outgroup taxa. In the past 30 years, there have been over 3000 publications dealing with some aspects of fleas, but only a few instances of formal cladistics analysis, so in-depth and continuous studies based on molecular data are needed to clarify the unknown phylogeny of this order. In addition, recent studies have demonstrated the existence of cryptic species, phenotypic plasticity or synonym taxa within the Siphonaptera. Fleas sometimes appear to have many instances of parallel morphological evolution, likely associated with multiple invasions of similar hosts. Even recent studies have provided incongruence between molecular and morphological results, emphasizing the necessity of combining morphological, phylogenetic and molecular data in order to assess and elucidate taxonomic issues regarding Siphonaptera.

Prof. Dr. Antonio Zurita, *Diversity* Guest Editor"

For more information and to submit manuscripts, see:
https://www.mdpi.com/journal/diversity/special_issues/8KDU234563

[This would be a great opportunity for flea experts to collaborate on flea-related articles. Editor]

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Lewis World Flea Species List

Updated 12 February 2022 and available on Flea News network on Entomological Society of America Networks

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Flea Presentation at the 8th International Congress of the Society for Vector Ecology

September 19-23, 2022, Honolulu, Hawaii,

"Mites, Lice, and Fleas, Oh My! A Regional Survey of Backyard Chickens",
Amy Murillo, University of California, Riverside.

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M.S. Thesis

Brittny N. Blakely (2022) Optimizing the Off-Host Production of Cat Fleas (Siphonaptera: Pulicidae) Using an Artifical Membrane-Based Blood-Feeding System. New Mexico St. Univ., Las Cruces.

Cat fleas are small blood feeding ectoparasites which feed on humans and animals. Their presence cause annoyance and discomfort through their bites and can transmit numerous diseases to both animals and humans. These diseases can include the plague, murine typhus, flea-borne spotted fever, and cat scratch disease; as well as an intestinal parasite, the double-pore tapeworm. Research on disease transmission, development of new anti-flea formulations, and monitoring insecticide resistance in field populations rely on the availability of large numbers of laboratory reared cat fleas. Traditionally, fleas are reared on live animals, but this requires animal handling permits, inflicts discomfort to the animals, and costs money and time to maintain the host animals. Although artificial membrane-based feeding systems have been proposed, these methods are not sustainable long-term because of reduced blood consumption and egg production which leads to fewer new adults than fleas reared on-host. This study optimizes the previously described artificial membrane-based feeding methods by comparing the blood consumption and egg production of cat fleas fed bloods from different species of host animal. This study also evaluates the effect on blood uptake of the addition of the phagostimulant adenosine triphosphate (ATP) to the bloods. Fleas fed dog blood had the highest blood consumption and egg production, but blood consumption did not increase when ATP was added to the bloods. Using dog blood, I raised one generation of cat fleas and observed a 610% growth rate in that generation. This is an improvement to the growth rates of cat fleas raised on cow blood using a similar feeding system that have been reported in literature. Developing a sustainable feeding method for rearing cat flea colonies without feeding on live animals will allow for a more humane and convenient production of this pest.

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Featured Research

New Flea Species 2022

(none so far)

Taxonomy

Berrizbeitia, M.F.L.
Incorrect neotype and neallotype designations for species of *Agastopsylla* (Siphonaptera: Ctenophthalmidae)
(2022) Zootaxa, 510.

Fleas of Argentina are receiving renewed systematic interest, but the identification of many species associated with small mammals can be problematic. We review the taxonomy of the flea genus *Agastopsylla* including the re-description of two species and one subspecies, and designate neotype and neallotype for *Agastopsylla hirsutior*, neotype for *Agastopsylla nylota nylota* from the “Colección

Mamíferos Lillo Anexos" (CMLA), Universidad Nacional de Tucumán, Argentina, and neotype and neallotype for *Agastopsylla pearsoni* from the Natural History Museum (London, U.K.). Additionally, a key to identification of the species of *Agastopsylla* and a distribution map of the species of the genus are included.

***

Zhang, Y., Fu, Y.-T., Yao, C., Deng, Y.-P., Nie, Y., Liu, G.-H.
Mitochondrial phylogenomics provides insights into the taxonomy and phylogeny of fleas
(2022) Parasites and Vectors, 15 (1), art. no. 223.

Background: Fleas (Insecta: Siphonaptera) are obligatory hematophagous ectoparasites of humans and animals and serve as vectors of many disease-causing agents. Despite past and current research efforts on fleas due to their medical and veterinary importance, correct identification and robust phylogenetic analysis of these ectoparasites have often proved challenging.

Methods: We decoded the complete mitochondrial (mt) genome of the human flea *Pulex irritans* and nearly complete mt genome of the dog flea *Ctenocephalides canis*, and subsequently used this information to reconstruct the phylogeny of fleas among Endopterygota insects.

Results: The complete mt genome of *P. irritans* was 20,337 bp, whereas the clearly sequenced coding region of the *C. canis* mt genome was 15,609 bp. Both mt genomes were found to contain 37 genes, including 13 protein-coding genes, 22 transfer RNA genes and two ribosomal RNA genes. The coding region of the *C. canis* mt genome was only 93.5% identical to that of the cat flea *C. felis*, unequivocally confirming that they are distinct species. Our phylogenomic analyses of the mt genomes showed a sister relationship between the order Siphonaptera and orders Diptera + Mecoptera + Megaloptera + Neuroptera and positively support the hypothesis that the fleas in the order Siphonaptera are monophyletic.

Conclusions: Our results demonstrate that the mt genomes of *P. irritans* and *C. canis* are different. The phylogenetic tree shows that fleas are monophyletic and strongly support an order-level objective. These mt genomes provide novel molecular markers for studying the taxonomy and phylogeny of fleas in the future.

***

Zurita, A., Rivero, J., García-Sánchez, Á.M., Callejón, R., Cutillas, C.
Morphological, molecular and phylogenetic characterization of *Leptopsylla segnis* and *Leptopsylla taschenbergi* (Siphonaptera)
(2022) Zoologica Scripta, 51 (6), pp. 741-754.

The taxonomic status of Leptopsyllidae family has remained controversial over the years. Thus, some entomologists placed this group of fleas within Ceratophyllidae family, considering it at level of Leptopsyllinae subfamily or even appearing as a paraphyletic group within Siphonaptera phylogeny. This fact is emphasized by the lack of molecular and phylogenetic data of Leptopsyllidae taxa available in public databases. The aim of this study was to carry out a comparative morphological, phylogenetic and molecular study of two species of *Leptopsylla* genus with zoonotic relevance (*Leptopsylla segnis* and *Leptopsylla taschenbergi*) isolated from rodents collected from different geographical areas of Europe in order to molecularly characterize both taxa and to establish their taxonomic and phylogenetic status within Leptopsyllidae family. For this purpose, we have analysed and compared several morphological traits between *L. segnis* and *L. taschenbergi* and compared five different molecular markers (ITS2, EF1-α, cox1, cox2 and cytb) among these both species and others belonging to Leptopsyllidae family. Based on the morphological results, we found a phenotypic plasticity phenomenon in one female specimen showing morphological characters of *L. segnis* but molecular sequences distinctive for *L. taschenbergi*. Furthermore, the molecular and phylogenetic analysis could easily discriminate among both species providing, by the first time, a monophyletic origin of Leptopsyllidae family. Lastly, with this work, we demonstrate one more time, the usefulness of the combination of mitochondrial and nuclear markers to solve taxonomic and phylogenetic issues within Siphonaptera field by the use of concatenated dataset.

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García-Sánchez, A.M., Zurita, A., Cutillas, C.
Morphometrics as a Complementary Tool in the Differentiation of Two Cosmopolitan Flea Species: *Ctenocephalides felis* and *Ctenocephalides canis*
(2022) Insects, 13 (8), art. no. 707.

Fleas (Siphonaptera) are one of the most important ectoparasites that represent a potential danger for the transmission of pathogens in our environment. The cat flea, *Ctenocephalides felis* (Bouché, 1835), and the dog flea, *Ctenocephalides canis* (Curtis, 1826) are among the most prevalent and most frequently studied species throughout the world. However, the variations observed in their morphological characteristics complicate their correct identification, especially when there is a lack of access to the equipment and funds required to carry out molecular biology techniques. With the objective to provide an additional tool to help in the differentiation of *Ctenocephalides* species, a principal component analysis was carried out for the first time in the present work on populations of *C. felis* and *C. canis* from countries in three continents, namely Spain (Europe), South Africa (Africa) and Iran (Asia). The factor maps assisted in the differentiation of both species and the detection of differences in overall size, although morphological ambiguity prevented the delimitation in populations of the same species. Thus, morphometrics represents a complementary tool to other traditional and modern techniques, with great potential to assist in the differentiation of fleas, particularly species that have historically been difficult to identify.

Zurita, A., García-Sánchez, Á. M., & Cutillas, C. (2022). Comparative molecular and morphological study of *Stenoponia tripectinata tripectinata* (Siphonaptera: Stenoponiidae) from the Canary Islands and Corsica. Bulletin of Entomological Research, doi:10.1017/S0007485322000098

Stenoponia tripectinata tripectinata (Tiraboschi, 1902) is the most prevalent subspecies, within the genus *Stenoponia*, in the Mediterranean area. This rodent flea is widely distributed throughout southwestern Europe and the North of Africa including Mediterranean islands and the Canary Islands.

Nevertheless, from a taxonomical and systematic point, this flea group has been neglected over the years. Therefore, the aim of this study was to carry out a comparative morphometric, phylogenetic, and molecular study of two populations of *S. t. tripectinata* isolated from rodents collected from different islands from the Canary Archipelago and from Corsica to clarify the taxonomic status of these two isolated populations and to assess the morphological and molecular differentiation between them. For this purpose, we have analyzed several morphological traits and sequenced five molecular markers (EF1-α, ITS2, cox1, cox2, and cytb). We observed slight differences in the overall body size between females of both populations, and two well-defined geographical genetic lineages. This suggests the existence of two cryptic subspecies within *S. t. tripectinata* corresponding to two different island groups. Furthermore, we bring to light the necessity to provide new and updated morphological, molecular, and phylogenetic data to clarify the taxonomic status of *S. tripectinata*.

Hosts and Distribution

Glebskiy, Y., Acosta-Gutiérrez, R. y Cano-Santana, Z. (2021) «*Plusaetis sibynus* (Siphonaptera: Ceratophyllidae): a new record of flea on *Didelphis virginiana*, with a checklist of fleas for this host», Therya Notes, 2, pp. 151-155. doi: 10.12933/therya_notes-21-53.

El tlacuache de Virginia (*Didelphis virginiana*) es hospedero para una variedad de pulgas que pueden ser importantes vectores para la transmisión de enfermedades, incluyendo zoonosis; por lo tanto, es importante generar conocimiento sobre las pulgas que parasitan a este animal. En este trabajo, presentamos la lista de pulgas que parasitan a los tlacuaches, con la adición de una especie que no ha sido registrada previamente en *D. virginiana*, ni en otros marsupiales. Colectamos pulgas de tlacuaches de Virginia en el área urbana de la Ciudad de México, México, y también realizamos una búsqueda bibliográfica para determinar las especies de pulgas reportadas previamente. Encontramos registros de 26 especies de pulgas en los tlacuaches y añadimos a esta lista la especie de pulga *Plusaetis sibynus*. *Plusaetis sibynus* es una pulga común en roedores (principalmente *Peromyscus* y *Neotoma*), por lo que representa un vector potencial para la transmisión de enfermedades de estos roedores a los tlacuaches, y posteriormente a las poblaciones humanas. Se requieren más estudios para determinar el papel de esta pulga como vector de enfermedades zoonóticas.

[The Virginia opossum (*Didelphis virginiana*) is host to a variety of fleas that can be important vectors for the transmission of diseases, including zoonoses; therefore, it is important to generate knowledge about the fleas that parasitize this animal. In this work, we present the list of fleas that parasitize opossums, with the addition of a species that has not been previously recorded in *D. virginiana*, nor in other marsupials. We collected fleas from Virginia opossums in the urban area of Mexico City, Mexico, and also conducted a literature search to determine previously reported flea species. We found records of 26 flea species on opossums and added the flea species *Plusaetis sibynus* to this list. *Plusaetis sibynus* is a common flea in rodents (mainly *Peromyscus* and *Neotoma*), which is why it represents a potential vector for the transmission of diseases from these rodents to opossums, and subsequently to human populations. Further studies are required to determine the role of this flea as a vector of zoonotic diseases.]

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Glebskiy, Yury, Roxana Acosta-Gutiérrez, Zenón Cano-Santana, Effect of urbanization on the opossum

Didelphis virginiana health and implications for zoonotic diseases, Journal of Urban Ecology, Volume 8, Issue 1, 2022, juac015, https://doi.org/10.1093/jue/juac01

Urban animals can be an important threat to human health as possible hosts of zoonotic diseases and their susceptibility to these diseases can depend on their overall health conditions. Thus, it is important to understand the factors that determine their health conditions. For this, we studied Virginia opossums (*Didelphis virginiana*) in six locations with different urbanization levels and types in Mexico City, Mexico. We trapped opossums and measured eight health-related characteristics (number of ectoparasites and tartar severity, among others) and estimated the percentage of area covered by the four main types of terrain (natural vegetation, managed vegetation, impermeable terrain and constructions). Data were analyzed by a canonical correspondence analysis. We found that impermeable terrain was related to negative health characteristics, while the constructions were opposite to impermeable terrain and mostly related to good health characteristics. At the same, time constructed areas held a smaller population than the natural areas. This suggests that constructed areas provide few shelters, but opossums are healthier there, while impermeable areas provide more shelter but cause more health problems to the animals, thus increasing the chances of zoonotic diseases. We recommend reducing the impermeable areas in city planning to contribute to a better health of the urban animals and therefore reduce risks of zoonotic diseases with potentially disastrous results.

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M. Fernanda López-Berrizbeitia, Roxana Acosta-Gutiérrez, M. Mónica Díaz.
Fleas of mammals and patterns of distributional congruence in northwestern Argentina: A preliminary biogeographic analysis.
Heliyon, 6: e04871 https://doi.org/10.1016/j. heliyon.2020.e04871

In few groups of parasites have the patterns of distribution been studied using quantitative methods, even though the study of these organisms indirectly provides information on the biogeographic history of their hosts, and in turn, the history of the hosts allows elucidation of speciation events of the parasites. Our objective was to quantitatively identify distributional congruence patterns of native fleas in northwestern Argentina. We analyzed 159 georeferenced distributional records of 47 species and six subspecies of fleas in northwestern Argentina using NDM/VNDM software. We found eight consensus areas, defined by 17 species and two subspecies, included in six patterns of distributional congruence (PDCs) with endemic and non-endemic fleas. The PDCs with the greatest values of endemicity (E) were mainly associated with Monte and Yungas Forests areas. All patterns indicated strong tendency of the Yungas Forests as a possible endemism area. Our results indicate that distributional congruence centers are generally located in Yungas Forests areas and highlight the importance of these areas in conservation and historical biology. This new information will allow delimitation of areas in the region at a more detailed resolution in the future.

Acosta, R., C. Guzmán-Cornejo, F. A. Quiñonez Cisneros, A. A. Torres Quiñonez, and J. A. Fernández. 2020. New records of ectoparasites for Mexico and their prevalence in the montane shrew *Sorex monticolus* (Eulipotyphla: Soricidae) at Cerro del Mohinora, Sierra Madre Occidental of Chihuahua, Mexico. Zootaxa, 4809(2): 393-396. https://doi.org/10.11646/ zootaxa.4809.2.11

The Flora and Fauna Protection Area (Área de Protección de Flora y Fauna—ÁPFF) Cerro del Mohinora, is the highest mountain in northern Mexico, reaching an elevation of 3,300 meters. It constitutes one of the last high-elevation islands of alpine and subalpine vegetation known in the Sierra Madre Occidental, in the extreme southwestern part of Chihuahua. The ÁPFF Cerro del Mohinora is located near the state border and limits with Durango and Sinaloa. This type of ecosystem located at high altitudes is in danger of disappearing since only 1% or less of its original extension remains; it is considered a refuge for species with boreal affinities (McDonald et al. 2011).

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Orlova, M.V., Larchanka, A.I., Dolgova, I.G., Dziamianchyk, V.V.
Unusual findings of fleas (Siphonaptera: Ctenophthalmidae, Ceratophyllidae) on bats (Chiroptera: Vespertilionidae) in Belarus: case report
(2022) Ecologica Montenegrina, 57, pp. 37-43. doi:10.37828/em.2022.57.5

Two fleas (*Paleopsylla soricis soricis* (Dale, 1878) (Siphonaptera: Ctenophthalmidae) and *Amalaraeus penicilliger* Grube, 1851 (Siphonaptera: Ceratophyllidae)) have been recorded on vesper bats (Chiroptera: Vespertilionidae) for the first time. Registration of *Am. penicilliger* in Belarus is the first geographical report.

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Ivanova-Aleksandrova, N., Dundarova, H., Neov, B., Emilova, R., Georgieva, I., Antova, R., Kirov, K., Pikula, J., Zukalová, K., Zukal, J.
Ectoparasites of Cave-Dwelling Bat Species in Bulgaria
(2022) Proceedings of the Zoological Society, pp. 1-6. doi:10.1007/s12595-022-00451-4

The unique ecological conditions of Bulgarian caves allow them to be used by bats year-round, thereby providing an interesting model for research on host-parasite interactions, including the potential for transmission of different zoonotic pathogens. In this study, 142 cave-dwelling bats of thirteen species were caught in seven Bulgarian caves and examined for presence of ectoparasites. Bats were mist-netted at cave entrances between May 2020 and May 2021. All macroscopically visible ectoparasites were collected from each bat and stored separately in 96% ethanol. The greatest diversity of bat ticks, flies and fleas was observed on greater horseshoe bats *Rhinolophus ferrumequinum* and common bent-wing bats *Miniopterus schreibersii*. *Spinturnix myoti* was the dominant ectoparasite collected at almost all localities and in all bat species. There was no significant difference in parasite load or diversity between the four most abundant bat species, each being infested with two specimens of a single parasite species on average. Though Bulgarian caves are used year-round by a range of bat species, parasite load and diversity remain low during the hibernation and migration periods. Mixed clusters of bats allow for inter-specific transmission of otherwise species-specific ectoparasites.

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Stevens, L., Stekolnikov, A.A., Ueckermann, E.A., Horak, I.G., Matthee, S.
Diversity and distribution of ectoparasite taxa associated with *Micaelamys namaquensis* (Rodentia: Muridae), an opportunistic commensal rodent species in South Africa
(2022) Parasitology, 149 (9), pp. 1229-1248.

South Africa boasts a rich diversity of small mammals of which several are commensal and harbour parasites of zoonotic importance. However, limited information is available on the parasite diversity and distribution associated with rodents in South Africa. This is particularly relevant for *Micaelamys namaquensis* (Namaqua rock mouse), a regionally widespread and locally abundant species that is often commensal. To address the paucity of data, the aims of the study were to record the ectoparasite diversity associated with *M. namaquensis* and develop distribution maps of lice and mites associated with *M. namaquensis* and other rodents in South Africa. *Micaelamys namaquensis* individuals (n = 216) were obtained from 12 localities representing multiple biomes during 2017-2018. A total of 5591 ectoparasites representing 5 taxonomic groups-fleas, lice, mesostigmatid mites, chiggers and ticks was recorded. These consisted of at least 57 taxa of which ticks were the most speciose (20 taxa). Novel contributions include new host and locality data for several ectoparasite taxa and undescribed chigger species. Known vector species were recorded which included fleas (*Ctenocephalides felis*, *Dinopsyllus ellobius* and *Xenopsylla brasiliensis*) and ticks (*Haemaphysalis elliptica*, *Rhipicephalus appendiculatus* and *Rhipicephalus simus*). Locality records indicate within-taxon geographic differences between the 2 louse species and the 2 most abundant mite species. It is clear that *M. namaquensis* hosts a rich diversity of ectoparasite taxa and, as such, is an important rodent species to monitor in habitats where it occurs in close proximity to humans and domestic animals.

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Steventon, C., Harley, D., Wicker, L., Legione, A.R., Devlin, J.M., Hufschmid, J.
An assessment of ectoparasites across highland and lowland populations of Leadbeater's possum (*Gymnobelideus* leadbeateri): Implications for genetic rescue translocations
(2022) International Journal for Parasitology: Parasites and Wildlife, 18, pp. 152-156.

Leadbeater's possum (*Gymnobelideus leadbeateri*) is a nocturnal arboreal marsupial with a restricted range centered on the Victorian Central Highlands, south-eastern Australia. Most populations inhabit wet montane ash forest and subalpine woodland, with one notable exception - a small, outlying and genetically-distinct lowland population inhabiting swamp forest at Yellingbo, Victoria. The species has been listed as critically endangered since 2015. Translocations are the mainstay of critical genetic rescue and this study explores the ectoparasites that are 'along for the ride' during translocation activities. Ectoparasites (133 fleas, 15 ticks and 76 mites) were collected opportunistically from 24 Leadbeater's possum colonies during population monitoring and genetic sampling across the lowland and highland populations. The composition of the flea assemblage varied by habitat type. Significantly greater numbers of the general marsupial fleas *Acanthopsylla r. rothschildii*. and *Choristopsylla tristis* (as a proportion of total flea numbers) were detected in lowland habitats, compared to highland habitats (Fishers exact test, $P < 0.0001$). Two host-specific flea species, *Stephanocircus domrowi* and *Wurunjerria warnekei* were detected only on possums in highland habitats. As a proportion of total fleas this was significantly different to possums in lowland habitats (Fisher's exact test, $P = 0.0042$ and $P < 0.0001$, respectively). *Wurunjerria warnekei* was suspected to be extinct prior to this study. Ticks (*Ixodes tasmanii*, n = 15) and mites (*Haemdoelaps cleptus* [sic, *Haemolaelaps cleptus*?], n = 47 and *H. anticlea*, n = 29) have been identified in Leadbeater's possums historically. The possible causes of the different flea assemblages may be environmental/climatic, or due to the historic geographic division between highland and lowland animals. The planned translocations of highland individuals to lowland habitats will expose lowland individuals to novel species of previously exclusively highland fleas with unknown indirect consequences, thus careful monitoring will be required to manage any potential risks.

Physiology

André, M.R., Neupane, P., Lappin, M., Herrin, B., Smith, V., Williams, T.I., Collins, L., Bai, H., Jorge, G.L., Balbuena, T.S., Bradley, J., Maggi, R.G., Breitschwerdt, E.B.
Using Proteomic Approaches to Unravel the Response of *Ctenocephalides felis felis* to Blood Feeding and Infection With *Bartonella henselae*
(2022) Frontiers in Cellular and Infection Microbiology, 12, art. no. 828082.

Among the *Ctenocephalides felis felis*-borne pathogens, *Bartonella henselae*, the main aetiological agent of cat scratch disease (CSD), is of increasing comparative biomedical importance.

Despite the importance of *B. henselae* as an emergent pathogen, prevention of the diseases caused by this agent in cats, dogs and humans mostly relies on the use of ectoparasiticides. A vaccine targeting both flea fitness and pathogen competence is an attractive choice requiring the identification of flea proteins/metabolites with a dual effect. Even though recent developments in vector and pathogen -omics have advanced the understanding of the genetic factors and molecular pathways involved at the tick-pathogen interface, leading to discovery of candidate protective antigens, only a few studies have focused on the interaction between fleas and flea-borne pathogens.

Taking into account the period of time needed for *B. henselae* replication in flea digestive tract, the present study investigated flea-differentially abundant proteins (FDAP) in unfed fleas, fleas fed on uninfected cats, and fleas fed on *B. henselae*-infected cats at 24 hours and 9 days after the beginning of blood feeding. Proteomics approaches were designed and implemented to interrogate differentially expressed proteins, so as to gain a better understanding of proteomic changes associated with the initial *B. henselae* transmission period (24 hour timepoint) and a subsequent time point 9 days after blood ingestion and flea infection.

As a result, serine proteases, ribosomal proteins, proteasome subunit α-type, juvenile hormone epoxide hydrolase 1, vitellogenin C, allantoinase, phosphoenolpyruvate carboxykinase, succinic semialdehyde dehydrogenase, glycinamide ribotide transformylase, secreted salivary acid phosphatase had high abundance in response of *C. felis* blood feeding and/or infection by *B. henselae*. In contrast, high abundance of serpin-1, arginine kinase, ribosomal proteins, peritrophin-like protein, and FS-H/FSI antigen family member 3 was strongly associated with unfed cat fleas. Findings from this study provide insights into proteomic response of cat fleas to *B. henselae* infected and uninfected blood meal, as well as *C. felis* response to invading *B. henselae* over an infection time course, thus helping understand the complex interactions between cat fleas and *B. henselae* at protein levels.

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Corrigendum to: Bossard, R.L. 2021. Temperature Phenology of Fleas (Siphonaptera) on Rodents (Mammalia:Rodentia). Entomological Society of America annual meeting, October 31-November 3, 2021, Denver, Colorado USA (poster).

The conclusion that xeric flea communities did not show thermal niche partitioning was incorrect. In fact, all 5 flea communities including xeric ones showed evidence of thermal niche partitioning. R.L. Bossard

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Lu, S., Andersen, J.F., Bosio, C.F., Hinnebusch, B.J., Ribeiro, J.M.C.
Integrated analysis of the sialotranscriptome and sialoproteome of the rat flea *Xenopsylla cheopis* (2022) Journal of Proteomics, 254, art. no. 104476.

Over the last 20 years, advances in sequencing technologies paired with biochemical and structural studies have shed light on the unique pharmacological arsenal produced by the salivary glands of hematophagous arthropods that can target host hemostasis and immune response, favoring blood acquisition and, in several cases, enhancing pathogen transmission. Here we provide a deeper insight into *Xenopsylla cheopis* salivary gland contents pairing transcriptomic and proteomic approaches. Sequencing of 99 pairs of salivary glands from adult female *X. cheopis* yielded a total of 7432 coding sequences functionally classified into 25 classes, of which the secreted protein class was the largest. The translated transcripts also served as a reference database for the proteomic study, which identified peptides from 610 different proteins. Both approaches revealed that the acid phosphatase family is the most abundant salivary protein group from *X. cheopis*. Additionally, we report here novel sequences similar to the FS-H family, apyrases, odorant and hormone-binding proteins, antigen 5-like proteins, adenosine deaminases, peptidase inhibitors from different subfamilies, proteins rich in Glu, Gly, and Pro residues, and several potential secreted proteins with unknown function.

Significance: The rat flea *X. cheopis* is the main vector of *Yersinia pestis*, the etiological agent of the bubonic plague responsible for three major pandemics that marked human history and remains a burden to human health. In addition to *Y. pestis* fleas can also transmit other medically relevant pathogens including *Rickettsia* spp. and *Bartonella* spp. The studies of salivary proteins from other hematophagous vectors highlighted the importance of such molecules for blood acquisition and pathogen transmission. However, despite the historical and clinical importance of *X. cheopis* little is known regarding their salivary gland contents and potential activities. Here we provide a comprehensive analysis of *X. cheopis* salivary composition using next generation sequencing methods paired with LC-MS/MS analysis, revealing its unique composition compared to the sialomes of other blood-feeding arthropods, and highlighting the different pathways taken during the evolution of salivary gland concoctions. In the absence of the *X. cheopis* genome sequence, this work serves as an extended reference for the identification of potential pharmacological proteins and peptides present in flea saliva.

Evolution

Ilinsky, Y., Lapshina, V., Verzhutsky, D., Fedorova, Y., Medvedev, S.
Genetic Evidence of an Isolation Barrier between Flea Subspecies of *Citellophilus tesquorum* (Wagner, 1898) (Siphonaptera: Ceratophyllidae)
(2022) Insects, 13 (2), art. no. 126.

[abstract in Flea News 89 March 2022]

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Wagnon, G.S., Pletcher, O.M., Brown, C.R.
Change in beak overhangs of cliff swallows over 40 years: Partly a response to parasites?
(2022) PLoS ONE, 17 (2 Febuary), art. no. e0263422.

Some birds exhibit a maxillary overhang, in which the tip of the upper beak projects beyond the lower mandible and may curve downward. The overhang is thought to help control ectoparasites on the feathers. Little is known about the extent to which the maxillary overhang varies spatially or temporally within populations of the same species. The colonial cliff swallow (*Petrochelidon pyrrhonota*) has relatively recently shifted to almost exclusive use of artificial structures such as bridges and highway culverts for nesting and consequently has been exposed to higher levels of parasitism than on its ancestral cliff nesting sites. We examined whether increased ectoparasitism may have favored recent changes in the extent of the maxillary overhang. Using a specimen collection of cliff swallows from western Nebraska, USA, spanning 40 years and field data on live birds, we found that the extent of the maxillary overhang increased across years in a nonlinear way, peaking in the late 2000's, and varied inversely with cliff swallow colony size for unknown reasons. The number of fleas on nestling cliff swallows declined in general over this period. Those birds with perceptible overhangs had fewer swallow bugs on the outside of their nest, but they did not have higher nesting success than birds with no overhangs. The intraspecific variation in the maxillary overhang in cliff swallows was partly consistent with it having a functional role in combatting ectoparasites. The temporal increase in the extent of the overhang may be a response by cliff swallows to their relatively recent increased exposure to parasitism. Our results demonstrate that this avian morphological trait can change rapidly over time.

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Zhang, Y., Nie, Y., Li, L.-Y., Chen, S.-Y., Liu, G.-H., Liu, W.
Population genetics and genetic variation of *Ctenocephalides felis* and *Pulex irritans* in China by analysis of nuclear and mitochondrial genes
(2022) Parasites and Vectors, 15 (1), art. no. 266.

Background: Fleas are the most economically significant blood-feeding ectoparasites worldwide. *Ctenocephalides felis* and *Pulex irritans* can parasitize various animals closely related to humans and are of high veterinary significance.

Methods: In this study, 82 samples were collected from 7 provinces of China. Through studying the nuclear genes ITS1 and EF-1α and two different mitochondrial genes cox1 and cox2, the population genetics and genetic variation of *C. felis* and *P. irritans* in China were further investigated.

Results: The intraspecies differences between *C. felis* and *P. irritans* ranged from 0 to 3.9%. The interspecific variance in the EF-1α, cox1, and cox2 sequences was 8.2–18.3%, while the ITS1 sequence was 50.1–52.2%. High genetic diversity was observed in both *C. felis* and *P. irritans*, and the nucleotide diversity of cox1 was higher than that of cox2. Moderate gene flow was detected in the *C. felis* and *P. irritans* populations. Both species possessed many haplotypes, but the haplotype distribution was uneven. Fu's Fs and Tajima's D tests showed that *C. felis* and *P. irritans* experienced a bottleneck effect in Guangxi Zhuang Autonomous Region and Henan province. Evolutionary analysis suggested that *C. felis* may have two geographical lineages in China, while no multiple lineages of *P.irritans* were found.

Conclusions: Using sequence comparison and the construction of phylogenetic trees, we found a moderate amount of gene flow in the *C. felis* and *P. irritans* populations. Both species possessed many haplotypes, but the distribution of haplotypes varied among the provinces. Fu’s Fs and Tajima’s D tests indicated that both species had experienced a bottleneck effect in Guangxi and Henan provinces. Evolutionary analysis suggested that *C. felis* may have two geographical lineages in China, while no multiple lineages of *P. irritans* were found. This study will help better understand fleas' population genetics and evolutionary biology.

Ecology

Baláž, I., Bogdziewicz, M., Dziemian-Zwolak, S., Presti, C.L.O., Wróbel, A., Zduniak, M., Zwolak, R.
From trees to fleas: masting indirectly affects flea abundance on a rodent host
(2022) Integrative Zoology. doi:10.1111/1749-4877.12671

Mast seeding causes strong fluctuations in populations of forest animals. Thus, this phenomenon can be used as a natural experiment to examine how variation in host abundance affects parasite loads. We investigated fleas infesting yellow-necked mice in beech forest after 2 mast and 2 non-mast years. We tested 2 mutually exclusive scenarios: (1) as predicted by classical models of density-dependent transmission, an increase in host density will cause an increase in ectoparasite abundance (defined as the number of parasites per host), versus (2) an increase in host density will cause a decline in flea abundance (“dilution,” which is thought to occur when parasite population growth is slower than that of the host). In addition, we assessed whether masting alters the relationship between host traits (sex and body mass) and flea abundance. We found a hump-shaped relationship between host and flea abundance. Thus, the most basic predictions are too simple to describe ectoparasite dynamics in this system. In addition, masting modified seasonal dynamics of flea abundance, but did not affect the relationship between host traits and flea abundance (individuals with the highest body mass hosted the most fleas; after controlling for body mass, parasite abundance did not vary between sexes). Our results demonstrate that pulses of tree reproduction can indirectly, through changes in host densities, drive patterns of ectoparasite infestation.

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Krasnov, B.R., Shenbrot, G.I., Khokhlova, I.S.
Regional flea and host assemblages form biogeographic, but not ecological, clusters: evidence for a dispersal-based mechanism as a driver of species composition
(2022) Parasitology, 149 (11), pp. 1450-1459.

We used data on the species composition of regional assemblages of fleas and their small mammalian hosts from 6 biogeographic realms and applied a novel method of step-down factor analyses (SDFA) and cluster analyses to identify biogeographic (across the entire globe) and ecological (within a realm across the main terrestrial biomes) clusters of these assemblages. We found that, at the global scale, the clusters of regional assemblage loadings on SDFA axes reflected well the assemblage distribution, according to the biogeographic realms to which they belong. At the global scale, the cluster topology, corresponding to the biogeographic realms, was similar between flea and host assemblages, but the topology of subtrees within realm-specific clusters substantially differed between fleas and hosts. At the scale of biogeographic realms, the distribution of regional flea and host assemblages did not correspond to the predominant biome types. Assemblages with similar loadings on SDFA axes were often situated in different biomes and vice versa. The across-biome, within-realm distributions of flea vs host assemblages suggested weak congruence between these distributions. Our results indicate that dispersal is a predominant mechanism of flea and host community assembly across large regions.

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Tomás, A., Pereira da Fonseca, I., Valkenburg, T., Rebelo, M.T.
Parasite island syndromes in the context of nidicolous ectoparasites: Fleas (Insecta: Siphonaptera) in wild passerine birds from Azores Archipelago
(2022) *Parasitology International*, 89, art. no. 102564.

Island syndrome, previously established for isolation process of insular vertebrates' populations, have been adapted to insular parasites communities, termed parasite island syndromes. In this work, were studied for the first time the insular syndromes for nidicolous ectoparasites of the bird species, *Turdus merula, Sylvia atricapilla, Fringilla coelebs* and *Erithacus rubecula* from Azores and the mainland Portugal. Flea species were only recorded on Azorean birds, namely *Dasypsyllus gallinulae* and *Ctenocephalides felis felis*, known as not host-specific parasites. In the absence of shared flea species between mainland and islands birds, a comparison among our fleas prevalence to Azores Islands and mainland fleas prevalence, recorded to others European studies, showed that Azorean host populations undergo higher prevalence than the mainland one. This result was consistent with parasite island syndromes predictions recorded to ectoparasites, hippoboscid flies and chewing lice, that fleas have higher prevalence on the Azores Islands compared to mainland Portugal. However, our results provide a new perspective to parasite island syndromes assumptions, namely in the context of nidicolous ectoparasites that spend only brief periods on the hosts' body.

Pathology and Control

Cooke, B.D.
Fifty-year review: European rabbit fleas, *Spilopsyllus cuniculi* (Dale, 1878) (Siphonaptera: Pulicidae),

enhanced the efficacy of myxomatosis for controlling Australian rabbits
(2022) Wildlife Research, doi:10.1071/WR21154.

European rabbit fleas were released among Australian wild rabbits in the late 1960s to supplement mosquitoes as vectors of myxoma virus. Data from study sites across southern Australia in the 1960s and 1970s are reviewed to discern common elements of flea-borne myxomatosis epizootics and a simple model is proposed to explain how virus virulence and food quality interact to determine rabbit abundance. Low, stable populations of rabbits implied that, despite virus attenuation and increased rabbit disease resistance, flea-borne myxomatosis was extremely important in controlling rabbit populations. Despite the enhancement of myxomatosis, livestock producers benefitted little from the additional pasture because marketing difficulties were not conducive to industry growth. Native wildlife likely benefitted, nonetheless.

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Loeb, J. (2022) Flea and tick treatments need regulation. Veterinary Record, 191 (4), p. 140.

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Rousseau, J., Castro, A., Novo, T., Maia, C.
Dipylidium caninum in the twenty-first century: epidemiological studies and reported cases in companion animals and humans
(2022) Parasites and Vectors, 15 (1), art. no. 131.

Background: Dipilidiosis is a parasitic disease caused by the tapeworm *Dipylidium caninum*. Fleas and, less frequently, lice act as an intermediate host, and their ingestion is required for infection to occur. While the disease mainly affects domestic and wild carnivores, it is also considered a zoonotic disease, with most human cases reported in children. *Dipylidium caninum* is considered to be the most common tapeworm infesting companion animals, but dipilidosis in humans is rare. The aims of this review were to improve current understanding of the epidemiology of this parasitosis and its management by the medical and veterinary community.

Methods: A comprehensive review of the published literature during the last 21 years (2000–2021) on the epidemiology, clinical features, diagnosis, treatment and prevention measures of *D. caninum* infection and dipilidiosis in companion animals and humans was conducted.

Results: Using predefined eligibility criteria for a search of the published literature, we retrieved and screened 280 publications. Of these, 161 (141 epidemiological studies, 20 case reports [16 human cases]) were considered for inclusion in this review. This parasitosis is present worldwide; however, despite being the most frequent cestode infection in animals, it is often underdiagnosed using common coprological techniques. Its diagnosis in humans has also proved challenging, being frequently confused with pinworm infection, leading to inappropriate treatment and to the persistence of the disease over time. Prevention measures include control of ectoparasites in animals and the environment, as well as regular deworming of animals, most commonly with praziquantel.

Conclusions: The diagnosis of dipilidiosis remains challenging in both animals and humans, primarily due to the low sensitivity of the diagnostic methods currently available and a lack of knowledge of the morphological characteristics of the parasite. Although treatment with the appropriate anti-cestode compounds is well tolerated and results in resolution of the infection, indiscriminate use of these compounds may predispose to an increase in resistance. Given the worldwide distribution of this parasite, it is essential to act on several fronts, with a focus on health education for children and animal owners and the control of intermediate hosts, both in animals and in the surrounding environment.

Microbial Symbiotes

Kirillov, A.A., Kirillova, N.Y., Ruchin, A.B.
Parasites, Bacteria and Viruses of the Edible Dormouse *Glis glis* (Rodentia: Gliridae) in the Western Palaearctic
(2022) Diversity, 14 (7), art. no. 562.

An overview of the parasites, bacteria and viruses of *Glis glis* (Rodentia, Gliridae) inhabiting the Western Palearctic is given. A total of 85 articles published from 1895 to 2021 were reviewed and analysed in our study. According to the literature's data, 104 species associated with *G. glis* are recorded: 4 viruses, 8 Protozoa, 6 Cestoda, 6 Trematoda, 4 Nematoda, 1 Heteroptera, 2 Anoplura, 39 Siphonaptera and 34 Acari. The most studied group is ectoparasites. To a lesser extent, parasitic worms in *G. glis* were studied. There is very little data about the dormouse protozoans and viruses. The most studied parasites, viruses and protozoans of *G. glis* are in Germany, where 21 species were noted. The largest number of parasites was found in the dormouse in Russia (22), but of two groups only: helminths and ectoparasites. Only 20 out of 104 parasite species recorded in *G. glis* are host-specific. Most parasites (60 species) found in *G. glis* have a Palaearctic and cosmopolitan distribution. Three viruses, six species of protozoa and three helminths have veterinary and medical significance as potential pathogens of dangerous zoonoses. Also, many species of fleas, mites and ticks found on *G. glis* are vectors of a number of dangerous vector-borne diseases in humans and domestic and wild animals.

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Sridhar, R., Dittmar, K., Williams, H.M.
Using Surface Washing to Remove the Environmental Component from Flea Microbiome Analysis
(2022) Journal of Parasitology, 108 (3), pp. 245-253.

Microbial metabarcoding is a common method to study the biology of blood-feeding arthropods and identify patterns of potential pathogen transmission. Before DNA extraction, specimens are often surface washed to remove environmental contaminants. While surface washing is common, its effects on microbial diversity remain unclear. We characterized the microbiome of the flea species *Ceratophyllus idius*, an avian ectoparasite, and a potential vector of pathogens, using high-Throughput 16S rRNA sequencing. Half of the nests from which fleas were collected were subjected to an environmental manipulation in which nesting materials were periodically replaced. In a crossed study design we surface washed half of the flea samples from each environmental condition to produce 4 experimental conditions. Environmental manipulations resulted in significant differences in the diversity and structure of the flea microbiome, but these differences were unapparent when specimens were surface washed. Furthermore, differential abundance testing of the experimental groups revealed that surface washing predominantly affected the abundance of bacterial groups that are characterized as environmental contaminants. These findings suggest that environmental changes primarily affect the surface microbiome of arthropods and that surface washing is a useful tool to reduce the footprint of the external microbiome on analysis.

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Urdapilleta, M., Pech-May, A., Lamattina, D., Burgos, E.F., Balcazar, D.E., Ferrari, W.A.O., Lareschi, M., Salomón, O.D.
Ecology of fleas and their hosts in the trifinio of north-east Argentina: first detection of *Rickettsia asembonensis* in *Ctenocephalides felis felis* in Argentina
(2022) Medical and Veterinary Entomology, 36 (1), pp. 20-29.

Fleas are important in public health due to their role as parasites and vectors of pathogens, including *Rickettsia*. The aim of this study was to evaluate the diversity, abundance and prevalence of fleas and the presence of *Rickettsia* in the trifinio of north-east Argentina. Fleas from household and synanthropic animals were obtained from urban and periurban areas. They were taxonomically identified and samples of 227 fleas in 86 pools were analysed by polymerase chain reaction targeting the gltA and ompB genes of *Rickettsia* spp. The study revealed that *Ctenocephalides felis felis* was dominant on dogs, cats and opossums, with higher prevalence in the periurban area. The Shannon–Wiener and Morisita–Horn indices expressed differences in the diversity and similarity values of the absolute abundances of the species between the areas compared. DNA amplifications revealed 30.8% *C. f. felis* pools positive for *Rickettsia* spp. Phylogenetic analysis showed that the haplotype obtained was identical to *Rickettsia asembonensis* from Peru and Brazil. This is the first detection in Argentina of *R. asembonensis* that infects *C. f. felis*, and we emphasize the importance of conducting research from a 'One Health' perspective on the role of opossums and rodents in the integration of the transmission cycles of rickettsial bacteria.

Book review

Population Biology of Vector-Borne Diseases (1st Edition)

by John M. Drake, Michael Bonsall, and Michael Strand

OUP Oxford, Print ISBN 9780198853251, 0198853254, eText ISBN 9780192594648, 0192594648

"*Population Biology of Vector-Borne Diseases* is the first comprehensive survey of this rapidly developing field. The chapter topics provide an up-to-date presentation of classical concepts, reviews of emerging trends, synthesis of existing knowledge, and a prospective agenda for future research. The contributions offer authoritative and international perspectives from leading thinkers in the field. The dynamics of vector-borne diseases are far more intrinsically ecological compared with their directly transmitted equivalents. The environmental dependence of ectotherm vectors means that vector-borne pathogens are acutely sensitive to changing environmental conditions. Although perennially important vector-borne diseases such as malaria and dengue have deeply informed our understanding of vector-borne diseases, recent emerging viruses such as West Nile virus, Chikungunya virus, and Zika virus have generated new scientific questions and practical problems. The study of vector-borne disease has been a particularly rich source of ecological questions, while ecological theory has provided the conceptual tools for thinking about their evolution, transmission, and spatial extent. *Population Biology of Vector-Borne Diseases* is an advanced textbook suitable for graduate level students taking courses in vector biology, population ecology, evolutionary ecology, disease ecology, medical entomology, viral ecology/evolution, and parasitology, as well as providing a key reference for researchers across these fields."

Review from: https://www.vitalsource.com/en-ca/products/population-biology-of-vector-borne-diseases-v9780192594648

[Editor's note: fleas are scarcely mentioned in this text, but its areas of inquiry are applicable to fleas.]

Contents (excerpted) from: https://www.vitalsource.com/en-ca/products/population-biology-of-vector-borne-diseases-v9780192594648 …

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Fleas in Art and History

The Canterbury Tales and Other Poems (c. 1400)
by Geoffrey Chaucer
(b. c. 1340s, London, England, d. 25 October 1400)
("Edited for Popular Perusal" by David Laing Purves (1838-1873), printed c. 1881, William P. Nimmo Publ., Edinburgh, transcription with updated spelling and with diacritical marks removed).

[Note: Creation of the arts continued during the Black Death of the Middle Ages, but with only few, often oblique references to pestilence (Doubleday, S.R. 2022. After the Plague. Teaching Co., Chantilly VA). Perhaps the people then looked to arts for escape and pleasantry, and not grim reminder. Chaucer mentions pestilence a few times in his Canterbury Tales, including in this excerpt from the Prologue. - Flea News Editor].

"With us there was a DOCTOR OF PHYSIC;
In all this worlde was there none him like
To speak of physic, and of surgery:
For he was grounded in astronomy.
He kept his patient a full great deal
In houres by his magic natural.
Well could he fortune the ascendent
Of his images for his patient,.
He knew the cause of every malady,
Were it of cold, or hot, or moist, or dry,
And where engender'd, and of what humour.
He was a very perfect practisour
The cause y-know, and of his harm the root,
Anon he gave to the sick man his boot
Full ready had he his apothecaries,
To send his drugges and his lectuaries
For each of them made other for to win
Their friendship was not newe to begin
Well knew he the old Esculapius,
And Dioscorides, and eke Rufus;
Old Hippocras, Hali, and Gallien;
Serapion, Rasis, and Avicen;
Averrois, Damascene, and Constantin;
Bernard, and Gatisden, and Gilbertin.
Of his diet measurable was he,
For it was of no superfluity,
But of great nourishing, and digestible.
His study was but little on the Bible.
In sanguine and in perse he clad was all
Lined with taffeta, and with sendall.
And yet he was but easy of dispense:
He kept that he won in the pestilence.
For gold in physic is a cordial;
Therefore he loved gold in special."

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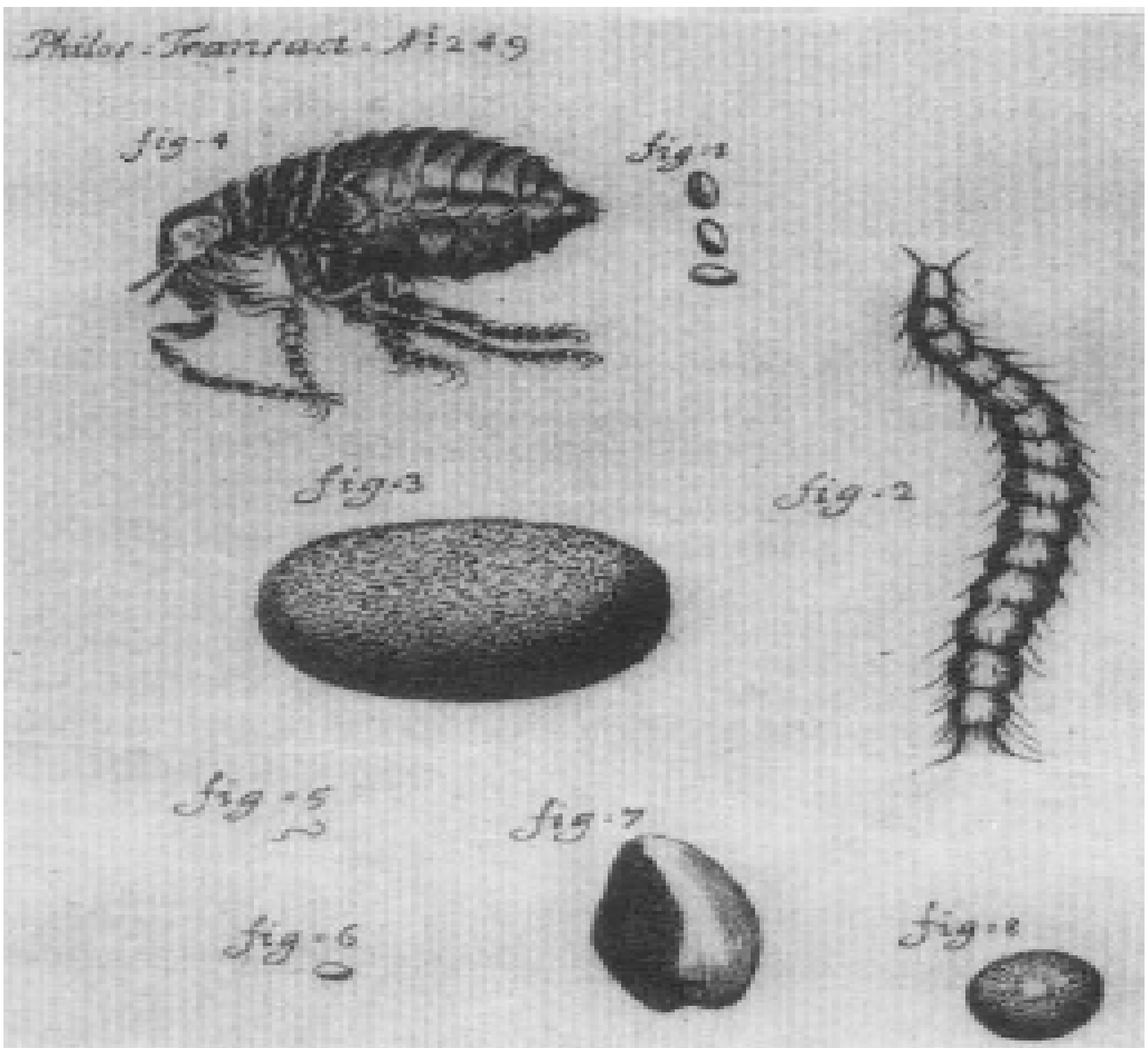
Cestoni, Giacinto (aka Diacinto)
(b. 13 May 1637, Montegiorgio, Italy, d. 29 January 1718, Livorno).
Life stages of a flea, drawing, 1699.
Phil. Trans. Royal Soc. 249.

[Cestoni observed the flea and scabies mite life cycles, also discovered aphid parthenogenesis (asexual reproduction) and viviparity (birth without external eggs), and collaborated with Francesco Redi, the experimenter who helped refute the hypothesis of spontaneous generation of flies from decaying meat.]



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